Psych Vocab Study Guide

131 terms from the notes. To quiz yourself, fold each page along the dashed line so only the terms show, then say the definition out loud.

Brain parts

BRAIN STEM	
Brain stem	Lowest part of the brain. Connects the spinal cord to higher regions and handles unconscious functions.
Medulla	Base of the brain stem. Controls involuntary, basic life functions: heart rate, breathing, sneezing, digestion and swallowing. Vital for life.
Pons	Regulates the sleep–wake cycle.
Midbrain	A relay system that transmits information for vision and hearing. Also plays a role in pain.
Reticular activating system	Similar to the brain's reward center. Controls some voluntary movement, eye movement, certain types of learning and cognition, and emotion.
CEREBRAL CORTEX	
Frontal lobe	The command center of the brain: decision making, judgement, problem solving, personality, reasoning, planning and language. Enables complex thoughts and hope for the future.
Parietal lobe	Processes body signals like pain, touch and temperature.
Motor cortex	Attached to the frontal lobe. Helps control voluntary movement.
Somatosensory cortex	Part of the parietal lobe. Holds the specific regions the parietal lobe takes signals from.
Temporal lobe	Near the ear. Helps process tone and melody, along with hearing and auditory memories.
Occipital lobe	The visual processing part of the brain: visuospatial processing, distance and depth perception, color, object and face recognition, and memory formation.
Broca's area	Essential for speech. Sends commands to the muscles to produce sounds.
Wernicke's area	Important for understanding speech, including your own speech.
HINDBRAIN	
Cerebellum	Responsible for muscle control, including balance and movement. Especially sensitive to alcohol.
CENTER OF THE BRAIN	
Corpus callosum	Lets the left and right hemispheres of the brain communicate.

LIMBIC SYSTEM	
Thalamus	The brain's sensory "switchboard." Relay station between the brain stem and the cortex that routes sensory information to where it needs to go.
Hippocampus	Stores long-term information: remembering facts and memories.
Hypothalamus	Survival drives and homeostasis. Alerts the body when it needs something and controls the pituitary gland.
Amygdala	Processes emotions and survival responses. Activated when you become afraid.

Brain chemicals

NEUROTRANSMITTERS	
Dopamine	Activates the reward and pleasure system, influencing motivation, decisions and more.
Serotonin	Inhibitory transmitter that regulates emotions and sleep.
Norepinephrine	Neurotransmitter and hormone that controls alertness, arousal and "fight or flight."
Glutamate	Excitatory neurotransmitter in the central nervous system. Makes an action potential more likely and helps long-term memory and learning.
GABA	Inhibitory neurotransmitter that slows down neural activity, calms the nervous system, and regulates mood, sleep and anxiety.
Endorphin	Produced in the central nervous system. Reduces pain and encourages pleasure and euphoria.
Substance P	Transmits pain signals from the peripheral nervous system to the central nervous system.
Acetylcholine	Enables muscle action, learning and memory, and overall cognitive functioning.

HORMONES	
Hormones	Chemical messengers that target specific parts of the body to cause change. Slower than neurotransmitters.
Adrenaline	Stress hormone from the adrenal gland, released in response to physical or mental stress, fear or excitement. Triggers "fight or flight."
Leptin	Protein hormone produced by fat cells that regulates hunger.
Ghrelin	Hunger hormone produced by an empty stomach that triggers hunger.
Melatonin	Sleep-inducing hormone made in the pineal gland. Regulates the body's 24-hour clock for sleep.
Oxytocin	Made by the hypothalamus and released by the pituitary gland. The biological basis for social bonding, trust and reproduction.

Neurons & nerves

NEURONS	
Neurons	The basic building blocks of the nervous system.
Glial cells	Non-neuron cells that protect and support neurons.
Reflex	An involuntary, unconscious movement in response to a stimulus, carried out by a fast neural pathway (the reflex arc).
Sensory neurons	Nerve cells in the peripheral nervous system that carry sensory info (eyes, ears, nose, etc.) to the central nervous system.
Motor neurons	Carry info from the central nervous system out to the body to activate muscles.
Interneurons	The "middlemen" in the central nervous system between sensory inputs and motor outputs.
Neural transmission	The electrochemical process neurons use to communicate within the brain and body.
Action potential	A brief, all-or-nothing electrical charge: the neuron either fires or it doesn't.
All-or-nothing principle	A neuron either fires completely or not at all. There is no in-between.
Threshold	The level of energy or stimulation needed to trigger a response.
Depolarization	A neuron becomes less negative than its resting state, making it more likely to reach an action potential.
Refractory period	The rest time right after an action potential, before the neuron can fire again.
Resting potential	When a neuron is negatively charged and not firing.
Reuptake	The sending neuron reabsorbs extra neurotransmitters from the synaptic gap.
Excitatory neurotransmitters	Chemical messengers that make a neuron more likely to fire an action potential.
Inhibitory neurotransmitters	Chemical messengers that make a neuron less likely to fire an action potential.
Multiple sclerosis	Chronic autoimmune disorder where the immune system attacks the protective layer (myelin) around nerve axons.
Myasthenia gravis	Chronic autoimmune disorder that blocks signals from nerves to muscles, causing muscle weakness.

NERVOUS SYSTEM	
Central nervous system	The brain and spinal cord. Makes decisions and sends out commands.
Peripheral nervous system	Everything outside the brain and spinal cord. Brings info to the central nervous system.

NERVOUS SYSTEM	
Autonomic nervous system	Division of the peripheral nervous system that controls involuntary body functions.
Sympathetic nervous system	Part of the autonomic nervous system that controls "fight or flight."
Parasympathetic nervous system	Part of the autonomic nervous system that controls "rest and digest."
Somatic nervous system	Division of the peripheral nervous system that controls voluntary movement.

Sensation

SENSATION BASICS	
Sensation	Input: the process by which sensory receptors and the nervous system detect raw energy from the environment.
Perception	Output: what the brain does with the sensory information it receives.
Transduction	Turning one kind of energy (light waves, sound waves, etc.) into neural signals the brain can process.
Absolute threshold	The minimum stimulus intensity a person can detect 50% of the time.
Difference threshold	The minimum difference between two stimuli that a person can notice 50% of the time. Also called the just-noticeable difference.
Weber's law	To be noticed as different, two stimuli must differ by a constant minimum percentage, not a constant amount.
Signal detection theory	Detecting a stimulus is not a fixed percentage. It also depends on experience, expectations and alertness.
Subliminal stimulation	Information presented below the level of conscious awareness (below the absolute threshold).
Sensory adaptation	Becoming less sensitive to a constant, unchanging stimulus after continued exposure, so it fades into the background.
Sensory interaction	One sense can influence or blend with another, like taste and smell.
Synesthesia	Neurological condition where one sense involuntarily activates another.

VISION	
Retina	Light-sensitive inner surface at the back of the eye. Contains the rods and cones that carry out visual transduction.
Blind spot	The point on the retina where the optic nerve leaves the eye, creating a gap in vision.

VISION	
Optic nerve	A bundle of nerve fibers that carries neural impulses from the eye to the brain.
Lens	The clear structure behind the pupil and iris that focuses incoming light rays for near and far objects.
Accommodation	The lens changing shape to focus on near or far objects. (In Piaget's theory, it means changing existing frameworks to fit new info.)
Nearsightedness	Close objects are clear but far objects are blurry, because light focuses in front of the retina.
Farsightedness	Near objects are blurry but far objects are clear, because light focuses behind the retina.
Photoreceptors	Specialized light-detecting sensory neurons in the retina that perform transduction.
Rods	Photoreceptors in the retina that detect black, white and grey.
Cones	Receptor cells in the retina that process detail and color. Come in red, green and blue types.
Young-Helmholtz trichromatic theory	The retina contains three types of cones: red, green and blue.
Opponent-process theory	Colors are processed in opposing pairs (red–green, blue–yellow, black–white), so you can't see a "reddish green."
Afterimages	After staring at a strong color, an image stays in your vision once it's gone, but in a different (opposite) color.
Ganglion cells	Specialized neurons whose fibers form the optic nerve.

HEARING	
Volley theory	Groups of neurons fire in a pattern that matches the frequency of a sound.
Frequency theory	A single neuron fires at a rate that matches the sound wave's frequency.
Place theory	Different sound frequencies stimulate different places in the ear (cochlea).
Sound localization	The auditory system's ability to identify where a sound is coming from.
Conduction deafness	Hearing loss from damage to the outer or middle part of the ear.
Sensorineural deafness	Hearing loss from damage to the inner part of the ear.

OTHER SENSES	
Olfaction	The sense of smell. Detects odor molecules to tell you how to react (smoke = run, sweet = happy). Uses the olfactory bulbs.
Gustation	The sense of taste, using taste buds: sweet, sour, salty, bitter, umami and oleogustus (fat). Supertasters have more taste buds.

OTHER SENSES	
Pheromones	Chemical substances released by the body to affect others.
Kinesthesis	The body's sense of its own position, weight and movement of body parts, so you don't run into things. Linked to the cerebellum.
Vestibular sense	The sense of balance and spatial orientation that keeps you upright. Linked to the cerebellum.
Tactile sense	The sense of touch, a mix of four senses: pain, pressure, cold and warmth. Processed in the somatosensory cortex.
Pain	A biopsychosocial experience: a harmful signal to the brain (biological), shaped by beliefs (psychological) and social norms (social).
Gate control theory	The spinal cord has a "neurological gate" that either blocks or allows pain signals to reach the brain.
Phantom limb sensation	Feeling sensations or pain in a limb that is no longer there.

Sleep

SLEEP	
REM sleep	A recurring, active sleep stage where dreams occur.
REM rebound	The body's tendency to get more REM sleep after being deprived of it.
Activation-synthesis theory	Theory that dreams are the brain trying to make sense of random neural activity during REM sleep.
Consolidation theory	Theory that sleep and dreams help turn fragile short-term memories into long-term memories.
Memory consolidation	The biological and cognitive process that turns short-term memories into long-term memories.
Narcolepsy	Chronic neurological sleep disorder where sleep suddenly happens during the daytime.
Sleep apnea	Disorder that causes breathing to stop briefly during sleep, with repeated momentary awakenings.
Somnambulism	The technical term for sleepwalking. Happens during NREM-3 sleep.

Drugs

HOW DRUGS WORK	
Psychoactive drugs	Chemicals that change behavior, perception, mood and consciousness by altering neurotransmitter activity at the synapse.

HOW DRUGS WORK	
Agonist	A molecule that binds to a receptor site and mimics or enhances a neurotransmitter's effect.
Antagonist	A molecule that binds to a receptor site and reduces or blocks a neurotransmitter's effect.
Reuptake inhibitors	Drugs that block the reabsorption of neurotransmitters.
Tolerance	A drug's effect gets weaker with regular use.
Addiction	Compulsive craving for something despite knowing the harmful consequences.
Withdrawal	Unpleasant physical, cognitive or emotional symptoms when cutting back on a substance.

STIMULANTS	
Stimulants	Drugs that excite neural activity and speed up body functions.
Caffeine	A stimulant that boosts alertness and body functions.
Cocaine	Addictive stimulant that triggers dopamine, serotonin and norepinephrine.

DEPRESSANTS & MORE	
Depressants	Drugs that reduce neural activity and slow down body functions.
Alcohol	A depressant that slows the central nervous system, impairing judgement, memory and physical movement.
Opioids	Drugs that act as agonists in the brain to reduce pain and cause euphoria.
Heroin	An opioid that acts as a depressant, relieving pain and causing euphoria.
Hallucinogens	Drugs that distort perception and cause sensory images that aren't real.
Marijuana	A mild hallucinogen that also releases dopamine.

Nature & nurture

NATURE & NURTURE	
Heredity	The genetic passing of characteristics from parents to offspring.
Nature	The idea that human behavior is shaped by biology.
Nurture	The idea that human behavior is shaped by life experiences.
Genetic predisposition	Being more likely to develop a trait because of inherited genes.
Evolutionary perspective	Explains human behavior as a product of natural selection.

NATURE & NURTURE	
Natural selection	Traits that help survival are passed on to the next generation and become more common than other traits.
Eugenics	The belief in "improving" the human population by controlling who has children.
Twin studies	Studying twins to see how much nature versus nurture shapes behavior.
Family studies	Studying families to see how inherited traits affect someone.
Adoption studies	Comparing adopted children's behavior with their adoptive and biological parents.